THE SOUND BOX

FABRICATION & COMPUTATIONAL SPECIFICATIONS

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ABSTRACT:
The Resonant Chamber challenges the traditional boundaries of minimal domestic space by treating architecture as a physical interface. Stripped of static ornamentation, the spatial volume acts as a hardware extension of a digital audio environment. By applying principles of Human-Computer Interaction (HCI) to structural materials, the pavilion utilizes Polymer Dispersed Liquid Crystal (PDLC) laminates and targeted thermal outputs to translate live, multi-band audio frequencies into immediate environmental states. The resulting enclosure is a tangible domestic representation of sound.

1. THE MATERIAL & ARCHITECTURAL STACK

The domestic footprint is compressed to an 8x8 meter volume, relying on an array of highly engineered materials to negotiate acoustics, privacy, and aesthetic warmth. The structural logic draws upon midcentury precedents, utilizing a dark-anodized steel mullion grid to support the active glazing.

At the center of the space, a Tacchini Sesann sofa anchors the human scale. Its continuous, cold tubular chrome framing physically binds the warm, oversized cushions, providing a tactile counterpoint to the sterile precision of the glass walls. To achieve pristine sonic fidelity within a highly reflective glass box, the ceiling plane is treated with a mathematically derived acoustic topology—a grid of downward-facing, high-density foam pyramids designed to aggressively scatter standing sound waves before they can return to the listener.

2. COMPUTATIONAL LOGIC & REACTIVITY PROTOCOLS

The monolith operates on a continuous feedback loop between the audio source and the structural hardware. Digital audio signals are split via a Fast Fourier Transform (FFT) algorithm into isolated frequency bins. These discrete data streams are then mapped to analog control voltages (0-60VAC) to drive the physical environment in real-time.

PROTOCOL 01: LOW-FREQUENCY VOLUMETRIC REACTIVITY

Target: 20Hz – 100Hz (Bass / Percussive Impact)
Actuator: Central Fire Pit & Ambient Lighting Array
Logic: Low-frequency amplitude is continuously analyzed and normalized against a moving average. When bass frequencies peak, the data triggers a surge in the gas regulator of the central fire pit. The physical flames exhibit rapid upward acceleration and volumetric expansion, while the ambient point-lighting dynamically scales in intensity. Linear interpolation (Lerping) is applied to the signal translation to ensure the thermal and visual decay feels organic rather than strobing.

PROTOCOL 02: HIGH-CONTRAST PDLC DYNAMICS

Target: 150Hz – 2kHz (Mid-Range / Acoustic / Vocals)
Actuator: Structural Glass Envelope (PDLC Film)
Logic: The exterior walls contain a specialized electrochromic film. When inactive, the liquid crystals scatter light, rendering the glass completely opaque and milky-white. The mid-range audio frequencies dictate the voltage applied to the busbars hidden in the ceiling plenum. As the music swells, voltage is cut: the glass aggressively loses its transmission properties and spikes in surface roughness. The architecture effectively ”closes in” on the user during intense sonic moments, providing a physical visualization of acoustic density.

3. SIMULATION & RENDERING ARCHITECTURE

To validate the spatial logic and reaction latency prior to physical fabrication, a full-stack, browser-based simulation was engineered using WebGL and the `Three.js` library.

While conventional architectural visualization relies on heavy, pre-computed static meshes (such as `.obj` or `.gltf` files) which introduce significant memory overhead and render-blocking latency, this prototype generates its environment dynamically at runtime. This object-oriented, algorithmic approach to 3D modeling closely mirrors the efficiency of instruction pipelining in computer architecture.

- **Texture Generation:** The complex acoustic ceiling avoids millions of polygons by utilizing a procedurally computed radial-gradient bump map, generated via an HTML5 Canvas API and mapped to the material's physical roughness.
- **Live Buffer Analysis:** The simulation does not use pre-baked animations. It leverages the Web Audio API to process live audio tracks. The main execution thread performs real-time FFT analysis 60 times per second, directly injecting the resulting frequency arrays into the rendering pipeline to physically alter the material state of the glass and the particle velocity of the fire.

By treating the architecture as an executable program, the simulation achieves a flawless 60 frames-per-second while maintaining physically correct lighting and dynamic material transitions, proving the viability of the ”Sound Box” as a highly responsive, interactive domestic machine.